# Supplemental Materials

# SAP–PVA manufactured-sand cementitious composites: performance, environmental-cost assessment, and empirical-model-guided neural prediction

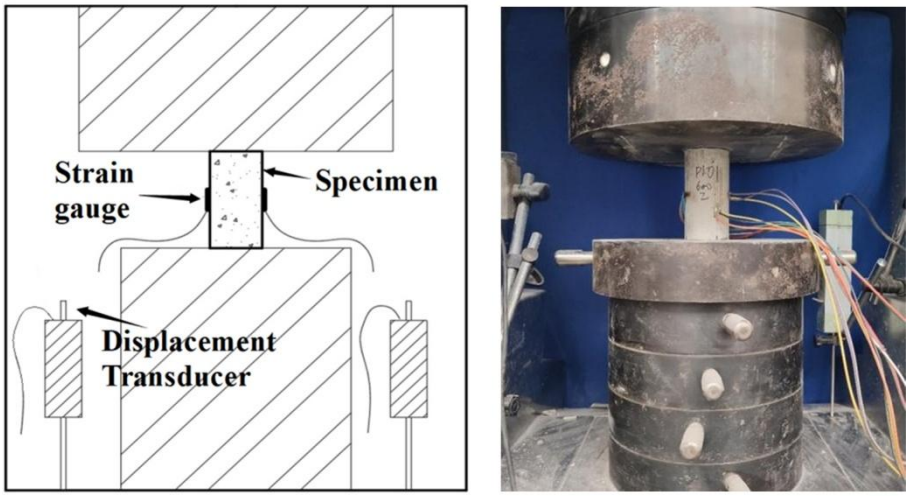
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## Supplementary Note 1. Figures




(a) Schematic diagram of the compressive test setup

(b) Loading device for the compressive test

**Figure S1.** Loading setup for compressive test

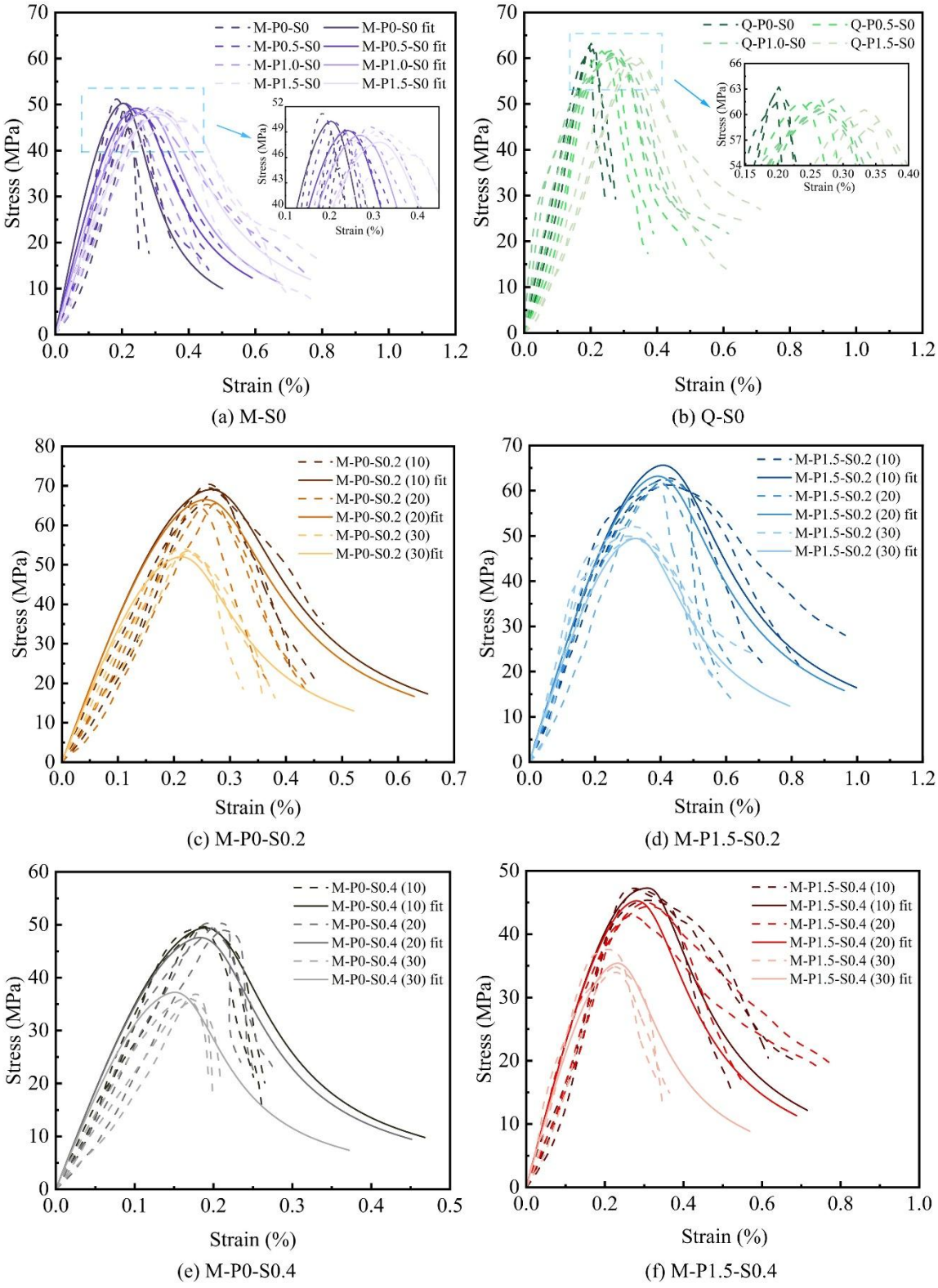



**Figure S2.** Comparison of compressive stress–strain curves and fitted curves of SAP-PVA-MSCC

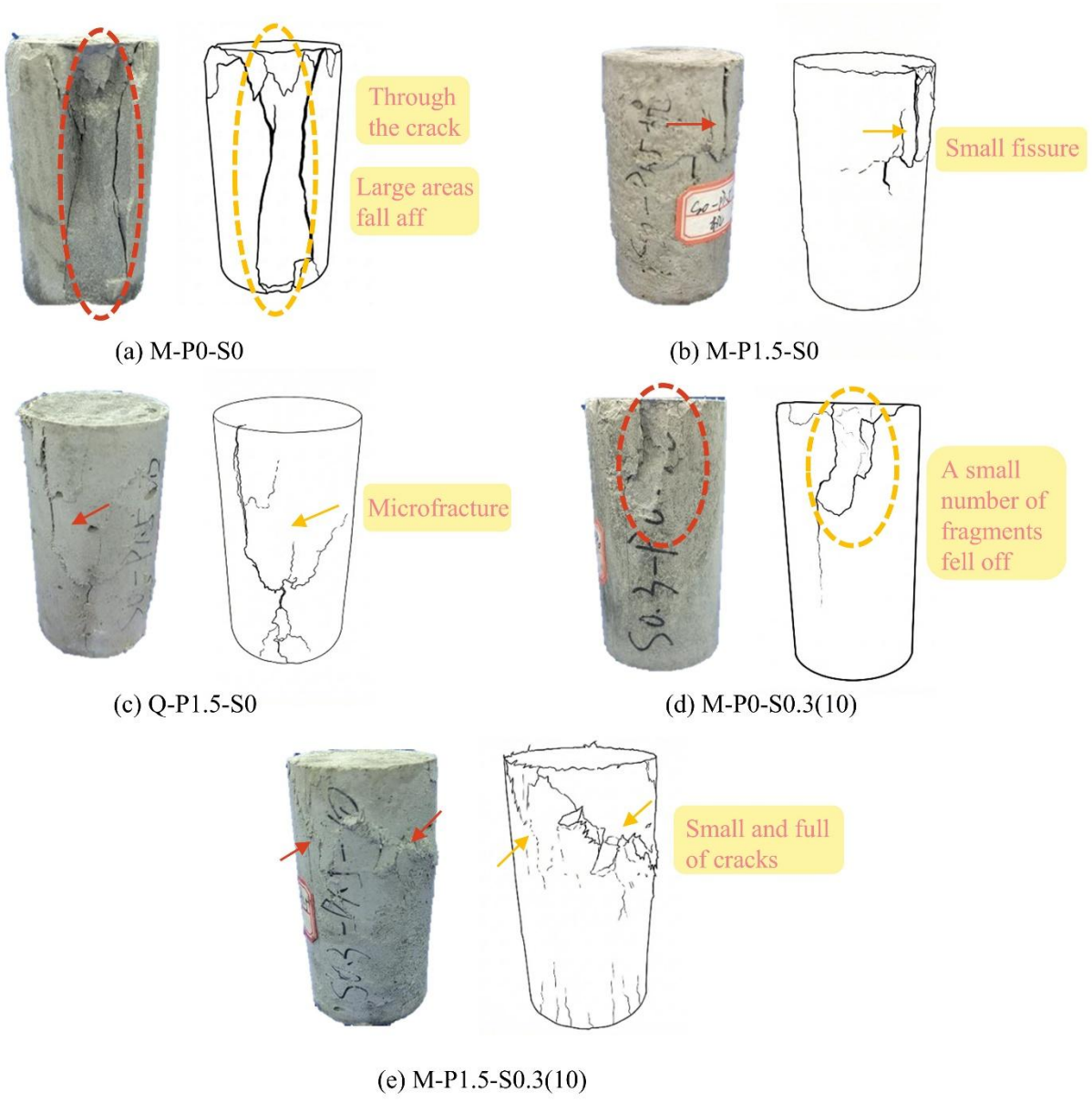



**Figure S3.** Typical compressive failure modes of SAP-PVA-MSCC specimens

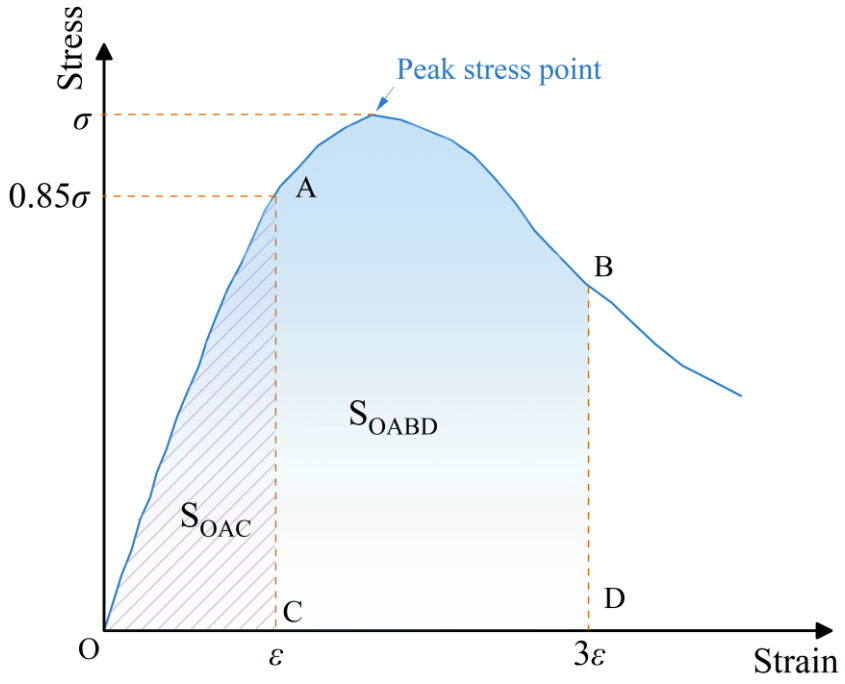



**Figure S4.** Schematic diagram of the calculation method for toughness index[1]

## Supplementary Note 2. Tables

**Table S1.** Physical properties and parameters of MIP test samples

| Specimen designation | Sample mass (g) | Bulk density (g/mL) | Apparent density (g/mL) |
|---|---|---|---|
| M-P1.5-S0.2(20) | 1.9597 | 1.6836 | 2.3095 |
| M-P1.5-S0.3(20) | 1.3601 | 1.7577 | 2.2615 |
| M-P1.5-S0.4(20) | 2.0143 | 1.6270 | 2.2908 |

**Table S2.** Summary of mechanical and chloride ion resistance performance indicators of different groups

| Specimen designation | Compressive strength (MPa) | Peak compressive strain (%) | Elastic modulus (GPa) | Toughness index | Total charge passed (C) |
|---|---|---|---|---|---|
| M-P0-S0 | 50.65 ± 0.91 | 0.209 ± 0.025 | 26.08 ± 1.56 | 2.44 ± 0.32 | 244 ± 5.56 |
| M-P0.5-S0 | 49.53 ± 1.30 | 0.237 ± 0.030 | 24.27 ± 1.63 | 3.95 ± 0.34 | 243 ± 4.13 |
| M-P1.0-S0 | 48.97 ± 1.14 | 0.274 ± 0.033 | 22.65 ± 1.23 | 4.12 ± 0.36 | 349 ± 6.89 |
| M-P1.5-S0 | 48.24 ± 1.56 | 0.328 ± 0.028 | 20.01 ± 1.21 | 4.50 ± 0.31 | 357 ± 6.54 |
| Q-P0-S0 | 62.60 ± 0.63 | 0.202 ± 0.006 | 32.10 ± 2.19 | 2.38 ± 0.32 | 201 ± 4.26 |
| Q-P0.5-S0 | 61.46 ± 0.33 | 0.254 ± 0.017 | 31.23 ± 1.47 | 3.53 ± 0.29 | 199 ± 4.87 |
| Q-P1.0-S0 | 61.04 ± 0.78 | 0.277 ± 0.010 | 29.45 ± 0.65 | 4.05 ± 0.37 | 250 ± 5.13 |
| Q-P1.5-S0 | 59.54 ± 0.89 | 0.336 ± 0.016 | 28.14 ± 0.91 | 4.31 ± 0.37 | 268 ± 6.48 |

| | | | | | |
|---|---|---|---|---|---|
| M-P0-S0.2(10) | 70.54 ± 0.67 | 0.267 ± 0.011 | 26.18 ± 2.21 | 3.28 ± 0.42 | 443 ± 7.36 |
| M-P0-S0.2(20) | 66.03 ± 0.16 | 0.258 ± 0.018 | 28.88 ± 1.10 | 2.72 ± 0.17 | 322 ± 5.49 |
| M-P0-S0.2(30) | 53.90 ± 0.73 | 0.220 ± 0.004 | 21.10 ± 2.02 | 3.39 ± 0.29 | 562 ± 7.38 |
| M-P0-S0.3(10) | 62.46 ± 0.88 | 0.263 ± 0.017 | 27.39 ± 1.54 | 4.83 ± 0.05 | 438 ± 4.26 |
| M-P0-S0.3(20) | 61.44 ± 2.66 | 0.223 ± 0.006 | 29.67 ± 1.13 | 2.79 ± 0.28 | 303 ± 2.79 |
| M-P0-S0.3(30) | 44.99 ± 0.95 | 0.204 ± 0.015 | 24.93 ± 2.04 | 5.15 ± 0.36 | 549 ± 7.59 |
| M-P0-S0.4(10) | 50.10 ± 0.31 | 0.187 ± 0.002 | 26.12 ± 1.19 | 2.52 ± 0.27 | 529 ± 7.17 |
| M-P0-S0.4(20) | 50.04 ± 0.84 | 0.206 ± 0.009 | 28.56 ± 1.60 | 2.48 ± 0.16 | 349 ± 5.68 |
| M-P0-S0.4(30) | 36.68 ± 0.75 | 0.172 ± 0.009 | 22.31 ± 1.68 | 2.69 ± 0.36 | 652 ± 8.04 |
| M-P1.5-S0.2(10) | 62.07 ± 0.89 | 0.418 ± 0.012 | 20.84 ± 1.84 | 2.97 ± 0.35 | 691 ± 7.65 |
| M-P1.5-S0.2(20) | 61.99 ± 0.75 | 0.398 ± 0.011 | 22.14 ± 1.49 | 2.82 ± 0.14 | 514 ± 6.62 |
| M-P1.5-S0.2(30) | 50.56 ± 1.36 | 0.306 ± 0.015 | 24.58 ± 2.00 | 2.73 ± 0.34 | 444 ± 5.82 |
| M-P1.5-S0.3(10) | 54.93 ± 1.22 | 0.369 ± 0.020 | 22.76 ± 1.27 | 2.86 ± 0.33 | 580 ± 6.41 |
| M-P1.5-S0.3(20) | 58.64 ± 0.49 | 0.362 ± 0.014 | 24.26 ± 1.85 | 2.17 ± 0.24 | 481 ± 5.20 |
| M-P1.5-S0.3(30) | 44.43 ± 1.28 | 0.320 ± 0.014 | 22.43 ± 1.88 | 2.02 ± 0.26 | 608 ± 6.71 |
| M-P1.5-S0.4(10) | 46.78 ± 0.95 | 0.284 ± 0.011 | 21.07 ± 2.14 | 5.10 ± 0.37 | 636 ± 6.32 |
| M-P1.5-S0.4(20) | 44.41 ± 1.15 | 0.294 ± 0.019 | 22.63 ± 2.20 | 4.96 ± 0.46 | 497 ± 4.64 |
| M-P1.5-S0.4(30) | 35.59 ± 1.89 | 0.227 ± 0.010 | 20.46 ± 1.82 | 3.75 ± 0.21 | 771 ± 9.16 |

**Table S3.** Carbon footprint, energy consumption, and cost under different mixture proportions

| Material | Q-P1.5-S0 | M-P0-S0 | M-P0-S0.2 | M-P1.5-S0 | M-P1.5-S0.2 | M-P1.5-S0.4 |
|---|---|---|---|---|---|---|
| A1-A3 Embodied carbon(kg $CO_2$-eq/$m^3$) | 628.11 | 566.00 | 575.69 | 622.94 | 632.64 | 637.70 |
| A1-A3 Embodied energy(MJ/$m^3$) | 5276.05 | 3276.10 | 3536.54 | 5234.20 | 5494.64 | 5660.46 |
| A1-A3 Material cost(CNY/$m^3$) | 5173.27 | 391.01 | 617.90 | 4884.04 | 5110.92 | 5261.42 |
| A4 Embodied carbon(kg $CO_2$-eq/$m^3$) | 218.52 | 144.83 | 144.79 | 142.72 | 142.68 | 142.29 |
| A4 Embodied energy(MJ/$m^3$) | 3586.98 | 2378.00 | 2377.33 | 2342.41 | 2341.74 | 2335.38 |
| A4 Material cost(CNY /$m^3$) | 530.09 | 349.97 | 349.87 | 346.93 | 346.83 | 345.89 |
| A5 Embodied carbon(kg $CO_2$-eq/$m^3$) | 0.63 | | | | | |
| A5 Embodied energy(MJ/$m^3$) | 2.66 | | | | | |
| A5 Material cost(CNY /$m^3$) | 0.53 | | | | | |
| Embodied carbon(kg $CO_2$-eq/$m^3$) | 847.26 | 711.46 | 721.11 | 766.29 | 775.95 | 780.62 |
| Embodied energy(MJ/$m^3$) | 8865.69 | 5656.76 | 5916.53 | 7579.27 | 7839.04 | 7998.5 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Material cost(CNY/$m^3$) | 5703.89 | 741.51 | 968.3 | 5231.50 | 5458.28 | 5607.84 |

**Table S4.** External LOSO-OOF prediction performance of compressive strength

| **Model** | $R^2$ **[95%CI]** | **RMSE/MPa[95%CI]** | **MAE/MPa** | **MAPE/%** | **Source-balanced RMSE/MPa** |
|---|---|---|---|---|---|
| EF | 0.8679[0.1867,0.9322] | 5.5553[3.7047,7.9508] | 3.9440 | 6.8093 | 5.2776 |
| BP-ANN | 0.7964[0.4523,0.8955] | 6.8977[3.6644,8.8401] | 5.3627 | 8.6708 | 4.6143 |
| JC-EF-R-BP | 0.9044[0.4852,0.9485] | 4.7264[2.8382,6.5720] | 3.3355 | 5.7393 | 4.1311 |

**Table S5.** External LOSO-OOF prediction performance of peak compressive strain

| **Model** | $R^2$**[95%CI]** | **RMSE/%[95%CI]** | **MAE/%** | **MAPE/%** | **Source-balanced RMSE/%** |
|---|---|---|---|---|---|
| EF | 0.8899[0.6798,0.9588] | 0.03104[0.02670,0.03583] | 0.02291 | 6.937 | 0.03213 |
| BP-ANN | 0.7407[0.3210,0.8765] | 0.04763[0.03702,0.06034] | 0.03910 | 13.837 | 0.04608 |
| JC-EF-R-BP | 0.9265[0.7561,0.9783] | 0.02536[0.02083,0.02750] | 0.02150 | 6.866 | 0.02235 |

## Supplementary Note 3. References

[1] J. Zhuang, et al. Study on the impact resistance of steel fiber-reinforced self-compacting concrete after high temperature, Structures 57 (2023) 105281.